

Hasan Al Reza, PhD (US Permanent Resident)

Memphis, TN | (513) 302-8641 | hasan.al.reza.bd@gmail.com | [LinkedIn](#) | [GitHub](#)

Experience

ST. JUDE CHILDREN'S RESEARCH HOSPITAL • Memphis, TN

Bioinformatics Research Scientist

Nov 2025 – Present

- Using **Multimodal NGS** (transcriptomics, epigenetics, spatial) to stage disease states and integrate genomic context to delineate the causal agents behind **neurological** and **hematological** disorders.
- Driving the advancement of **agentic AI skills and technologies** for scalable annotation, transcription factor footprinting, and biological interpretation of complex epigenetic datasets.

CINCINNATI CHILDREN'S HOSPITAL MEDICAL CENTER • Cincinnati, OH

Bioinformatics Data Scientist

Aug 2024 – Oct 2025

- Led the development of **Spatial Transcriptomic (Visium HD)** tools in analysis of pulmonary development & diseases with **10X higher resolution, leading to detection of 2 unique niches**.

Research Fellow

Nov 2023 – July 2024

- Spearheaded research into the development of Zonal Liver Organoids using **Molecular and Computational Biology** which has resulted in a 152% increase in survival in transplanted rats.
- Analyzed **NGS datasets using self-developed pipelines** to identify target genes and pathways which was used to identify 2 key transcription factors involved in zonation with 95% confidence.
- Performed **GWAS analysis using PLINK and GWAS tools** on multi-donor organoid library for studying MASH and characterized variants in GCKR and PNPLA3 genes associated with the disease.

Graduate Research Assistant

Jul 2018 – Oct 2023

- Led research into liver development, disease, and therapeutics using CRISPR edited stem cell derived HLO and Computational Biology that led to **1 patent and 4 research articles** in Q1 Journals.
- Created pipelines for analysis of **RNA-Seq, ATAC-seq, ChIP-Seq, and scRNA-Seq** datasets using latest memory unstrained bioinformatics algorithms that were used to analyze **Terabytes** of data.
- Analyzed **Perturb-seq and CROP-seq** data to identify the role of BSEP and PCK1 in metabolic dysfunction of the liver resulting in identification of **3 unique drug leads**.
- Increased productivity and accuracy by replacing manual cellular measurements with digital imaging technology using Python program reducing time constraints from **2 days to 2 hours**.
- Mentored a Medical Student in a research program who won the **1st prize** at the national level.

UNIVERSITY OF DHAKA • Dhaka, Bangladesh

Graduate Research Assistant

Mar 2016 – Oct 2023

- Developed potential **vaccines** and identified lncRNA and miRNA **biomarkers using genome wide analysis** for early disease detection which led to over 5 subsequent research articles in Q1 Journals.
- Managed and analyzed blood samples and **SNP data** collected from lung cancer patients at NICRH, BD resulting in **2 1st author publications** and introduction of techniques for early disease detection.
- Employed deep neural learning and other machine learning algorithms to predict mutations in **SARS-CoV-2** and accurately forecasted the change in viral genome by **17.4% over 2000 variants**.
- Assembled and analyzed **whole genome sequencing** data of MRSA samples from NICRH to identify **5 different strains** with multidrug resistance genes which caused 12.7% increase in hospital death.

Undergraduate Research Assistant

Jan 2014 – Feb 2016

- Identified SNPs in the Bangladeshi population that increased the risk of Type 2 Diabetes by 15%.
- Sequenced, analyzed, and submitted DNA barcodes for over 200 fishes and marine life of Bangladesh to the NCBI Nucleotide database resulting in identification of 2 new species.

Education

PhD in Molecular and Developmental Biology | University of Cincinnati

Oct 2023

MS in Genetic Engineering and Biotechnology | University of Dhaka

Sep 2017

BS in Genetic Engineering and Biotechnology | University of Dhaka

Feb 2016

Technical Skills

LANGUAGES: Python | SQL (PostgreSQL) | R | Bash (Unix/Linux) | JavaScript (npm) | HTML | VB.NET

BIOINFORMATICS, DATA SCIENCE & MACHINE LEARNING: Data Processing and Analysis (Pandas | Numpy | Scipy | Geopy) | Data Visualization (Matplotlib | Seaborn | Plotly | ggplot) | NLP | Deep Learning/Neural Network/LSTM (TensorFlow | Keras | PyTorch) | Linear & Logistic Regression/SVM/Decision Tree/Random Forest (Scikit-Learn) | Unsupervised and Supervised Learning (K-Means | Clustering | PCA | KNN) | Bioconductor | NGS Analysis (RNA-Seq | ChIP-Seq | scRNA-Seq) | AWS (Cloud Computing) | HPC (LSF) | Docker | Git | Nextflow | PLINK | GWASTools

DATA EXPERIENCE: Data Visualization | Process Optimization & Automation | Metabolomics | OMIM | GWAS | Modeling | Systems Biology | NGS | Mass Spectrometry | FLOW | CLIA/CAP | R Shiny | API

Computational Biology, Modeling & Machine Learning Projects

OMICS DATA ANALYSES:

- **RNA-Seq Analysis pipeline** – An end-to-end pipeline for bulk RNA-Seq data analysis using Salmon, Kallisto, Bowtie2, and Stringtie algorithms. The pipeline includes all the steps from quality control and alignment of FASTQ files to biological network analysis and functional enrichment analysis using external databases like GEO and KEGG. **Code:** [RNA-Seq](#) • **Languages:** R & Bash (Shell script).
- **ChIP-Seq Analysis pipeline** – Complete pipeline for analyzing ChIP-seq datasets using BWA and bowtie2 algorithms which includes quality control, mapping and differential analysis of genome wide binding sites cross-referenced with expression datasets. **Code:** [ChIP-Seq](#) • **Language:** R & Bash (Shell script).
- **scRNA-Seq Analysis pipeline** – Advanced pipeline utilizing the latest Alevin-Fry, STARsolo, and CellRanger algorithms for aligning single cell and single nuclear datasets. Pipeline also includes data analysis using Seurat, GSEA analysis, and functional enrichment using Cytoscape. **Code:** [scRNA-Seq](#) • **Language:** R, Python, Bash (Shell script) & LSF.

MACHINE LEARNING:

- **Deep Neural Learning** – An artificial Recurrent Neural Network (RNN) called Long Short Term Memory (LSTM) Network used to build a mutation prediction model for the next 2000 potential SARS-CoV-2 variants. **Code:** [Machine Learning](#) • **Language:** HTML & Python.

Selected Patents and Publications

Peer Reviewed Research Article:  [Google Scholar](#)

- **Reza HA**, Takebe T. Liver Organoid Model For Hyperbilirubinemia And Methods Of Making And Using Same. **WO/2022/261471 | PCT/US2022/033066**. 2022 Dec 15.
- Chen K, Zhao S, ... , **Reza HA**, Wagner A, Cakar AC, Jiang C, Zhang E, Green J, Martin E, Wikenheiser-Brokamp K. Decoding lymphangioliomyomatosis (LAM) niche environment via integrative analysis of single cell multiomics and spatial transcriptomics. *bioRxiv*. 2025 Jul 10:2025-07.
- **Reza HA**, Santangelo C, Iwasawa K, Reza AA, Sekiya S, Glaser K, Bondoc A, Merola J, Takebe T. Multi-zonal liver organoids from human pluripotent stem cells. *Nature*. 2025 Apr 16:1-0.
- **Reza HA**, Farooqui Z, ... , Takebe T (2023). Synthetic augmentation of bilirubin metabolism in human pluripotent stem cell-derived liver organoids. *Stem Cell Reports*. 2023 Nov 14;18(11):2071-83.
- Tonmoy MI, Fariha A, ... , **Reza HA**, Bahadur NM, Hossain MS. Computational epigenetic landscape analysis reveals association of CACNA1G-AS1, F11-AS1, NNT-AS1, and MSC-AS1 lncRNAs in prostate cancer progression through aberrant methylation. *Scientific Reports*. 2022 Jun 17;12(1):10260.
- **Reza HA**, Okabe R, Takebe T. Organoid transplant approaches for the liver. *Transplant International*. 2021;34(11):2031-45.
- Hossain MS, Pathan AS, ... , **Reza HA**, Roy M, Bahadur NM. Genome-wide identification and prediction of SARS-CoV-2 mutations show an abundance of variants: Integrated study of bioinformatics and deep neural learning. *Informatics in Medicine Unlocked*. 2021;27:100798.